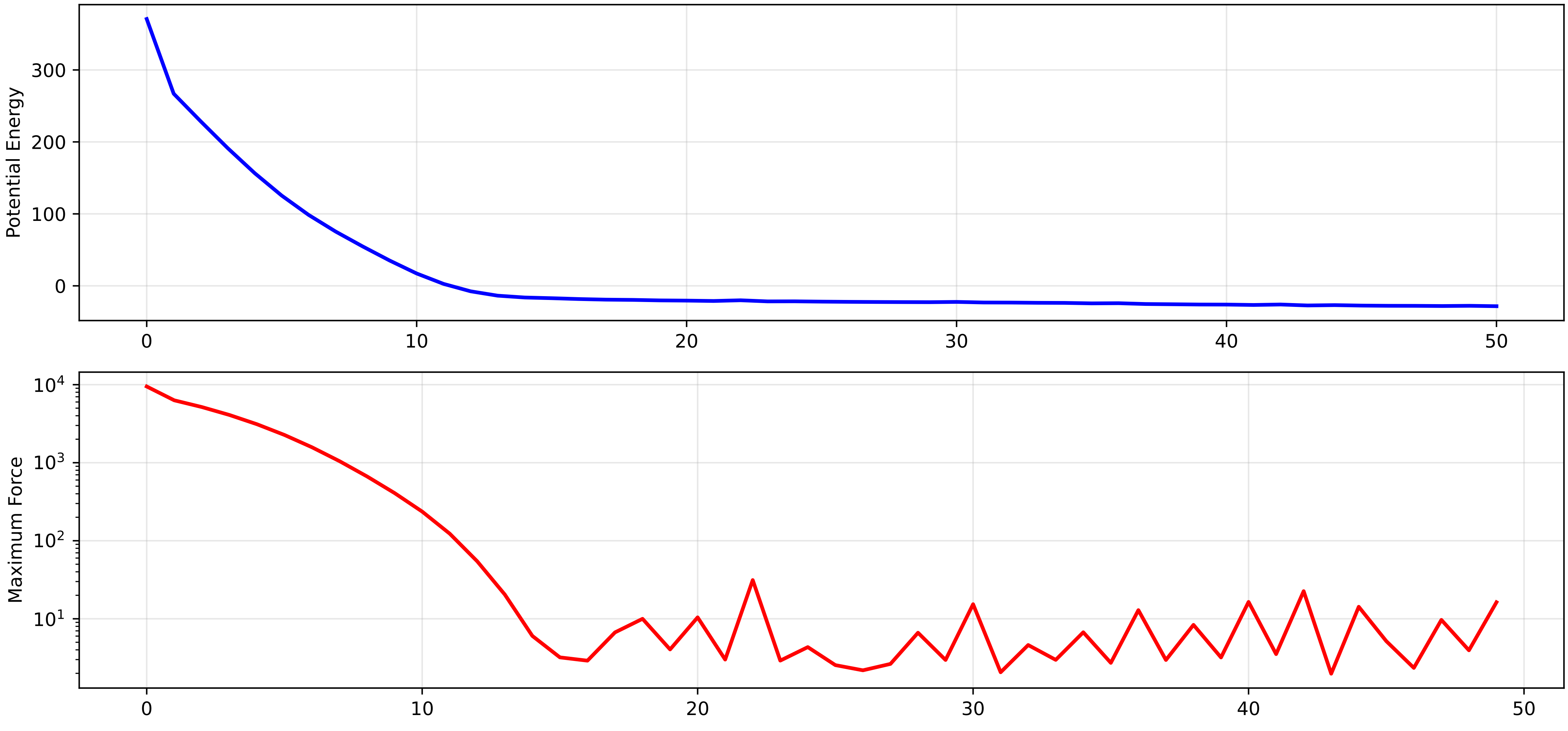
Energy Minimization Process in Molecular Dynamics



Energy Minimization in Molecular Dynamics:

1. Initial configuration: System starts in a high-energy, non-equilibrium state with overlapping particles or unfavorable interactions.

2. Steepest Descent Method: Forces on each atom are calculated from the gradient of the potential energy function. Atoms are moved in the direction of the force.

3. Process continues iteratively until:
   - Maximum force falls below tolerance (convergence criterion)
   - Maximum number of steps is reached

4. Step size adaptation:
   - Increased when energy decreases (faster convergence)
   - Decreased when energy increases (finer adjustments)

Applications:
- Removing steric clashes in initial structures
- Relaxing systems before dynamics
- Preparing systems for equilibration and production runs